

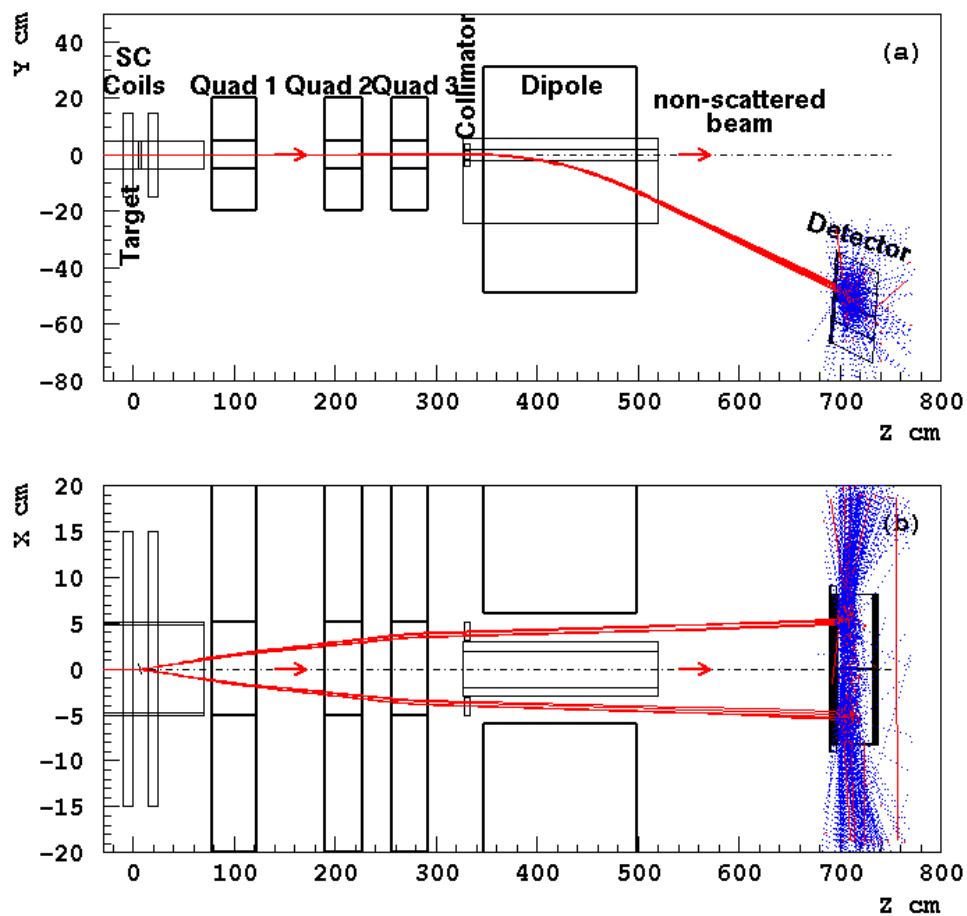
Møller Polarimeter Control Screens

EDM menu paths

9 March 2010

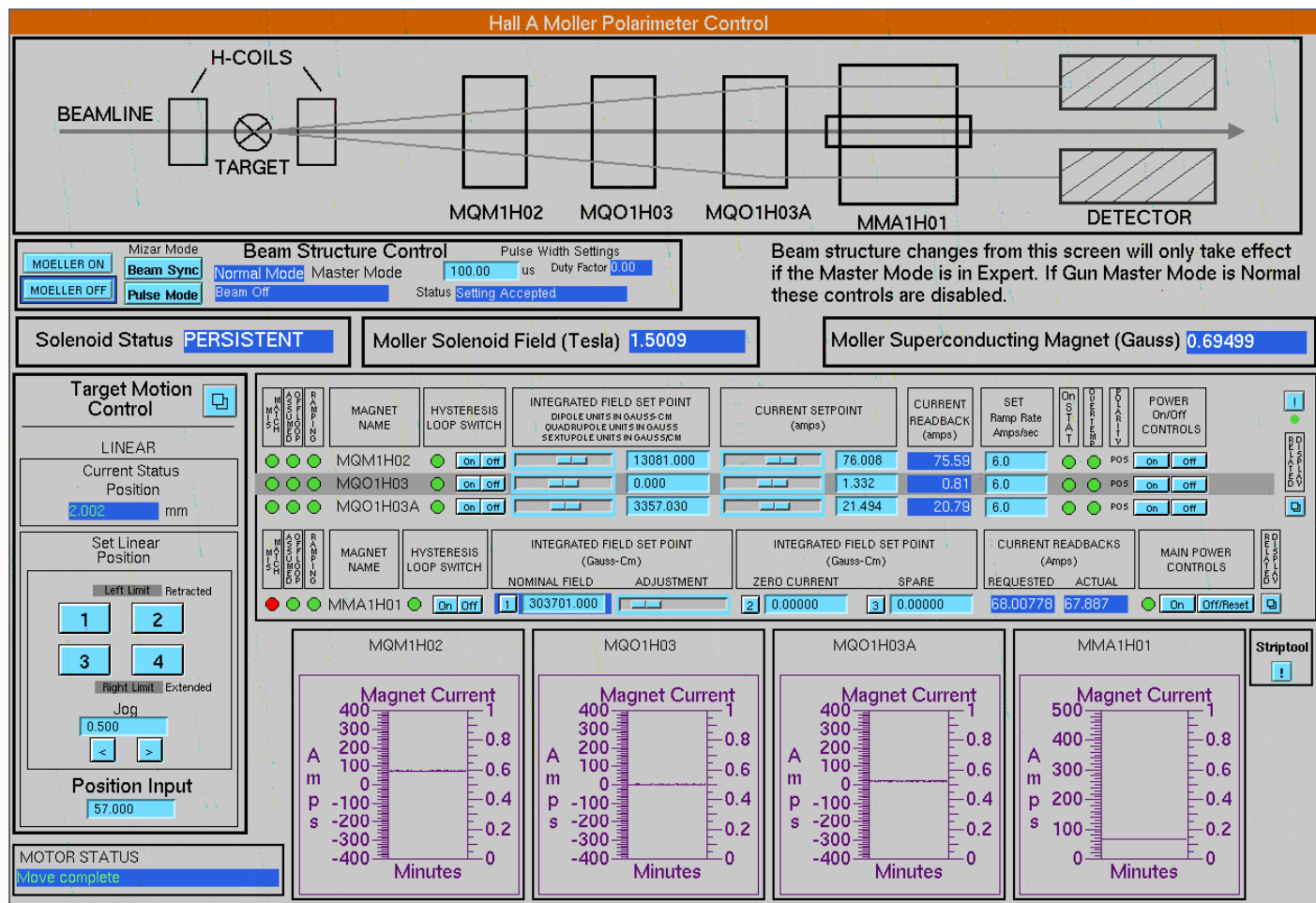
This is the 2010 EDM-menu version. The current 2016 JTabs / MyMenu screens are on the polarimeter control screens page.

The Møller Polarimeter Layout



The Møller Polarimeter Main Screen

EDM(OPS) → Acc.Main Menu → Hall A → Hall A Moller Polarimeter Control



The Møller Target Control Screen

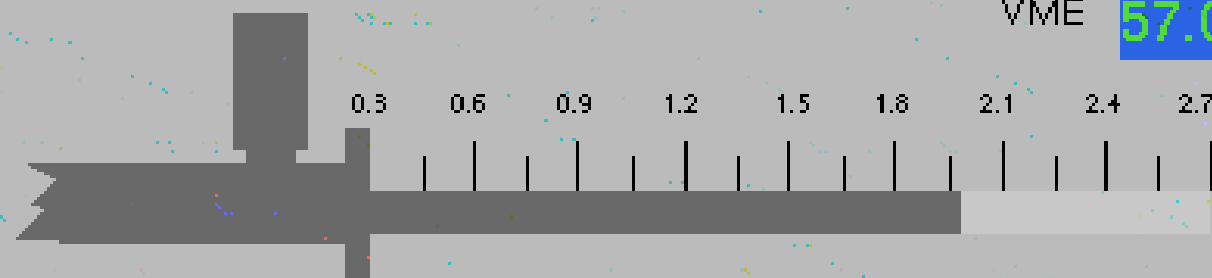
EDM(OPS) → Acc.Main Menu → Hall A → Hall A Moller Polarimeter Control → Moller Motion Control

Hall A Moller Target



Linear

Encoder **2.002**
VME **57.000**



LL Retracted

Extended RL

Controls

Retract

PARK

1

1

2

2

3

3

4

4

57.000

<

0.500

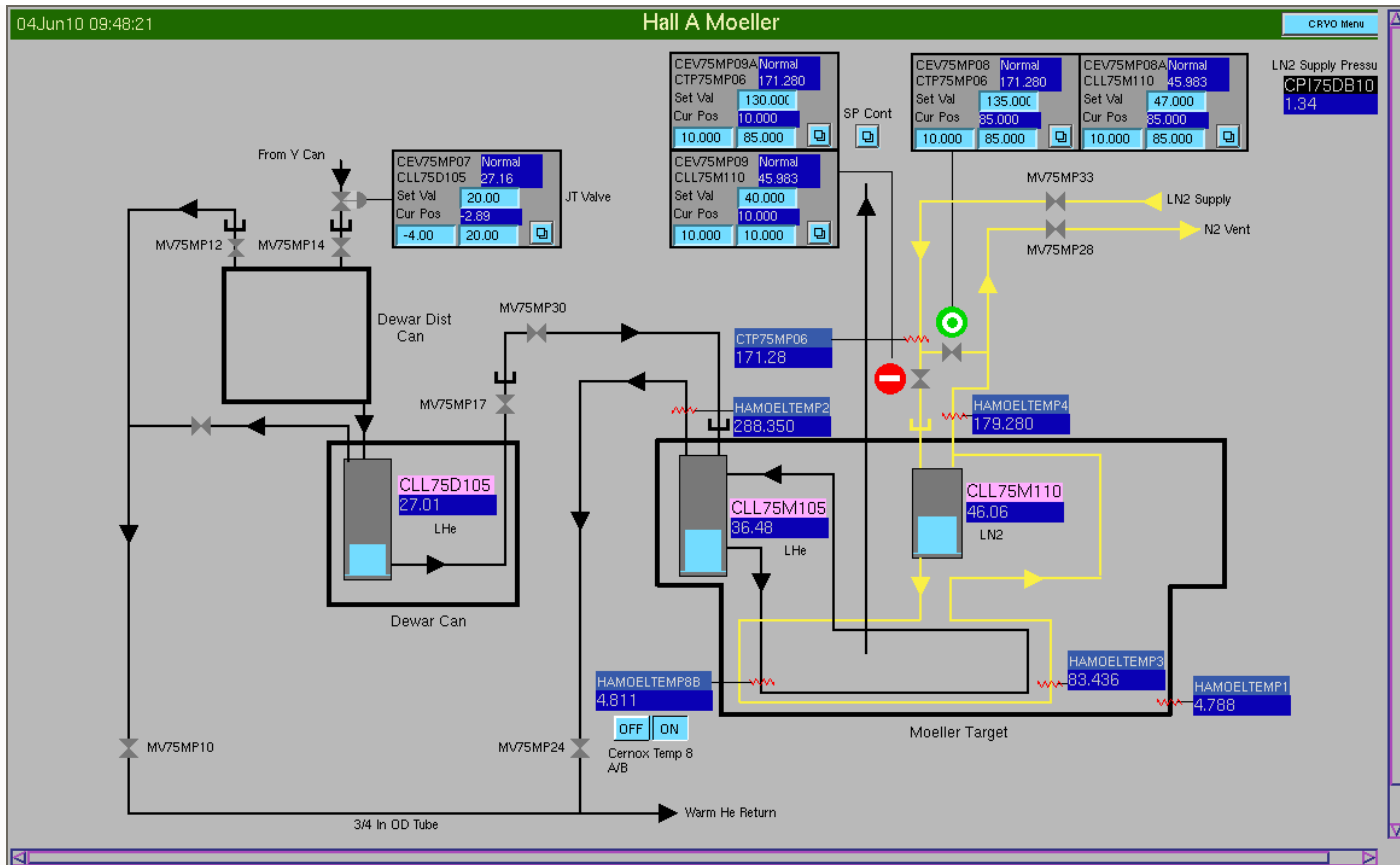
>

STOP

Move complete

The Møller Magnet Cryo Control Screen

EDM(Cryo Main) → ESR → Hall A Menu → Hall A - DB/Matrix → Hall A Moller

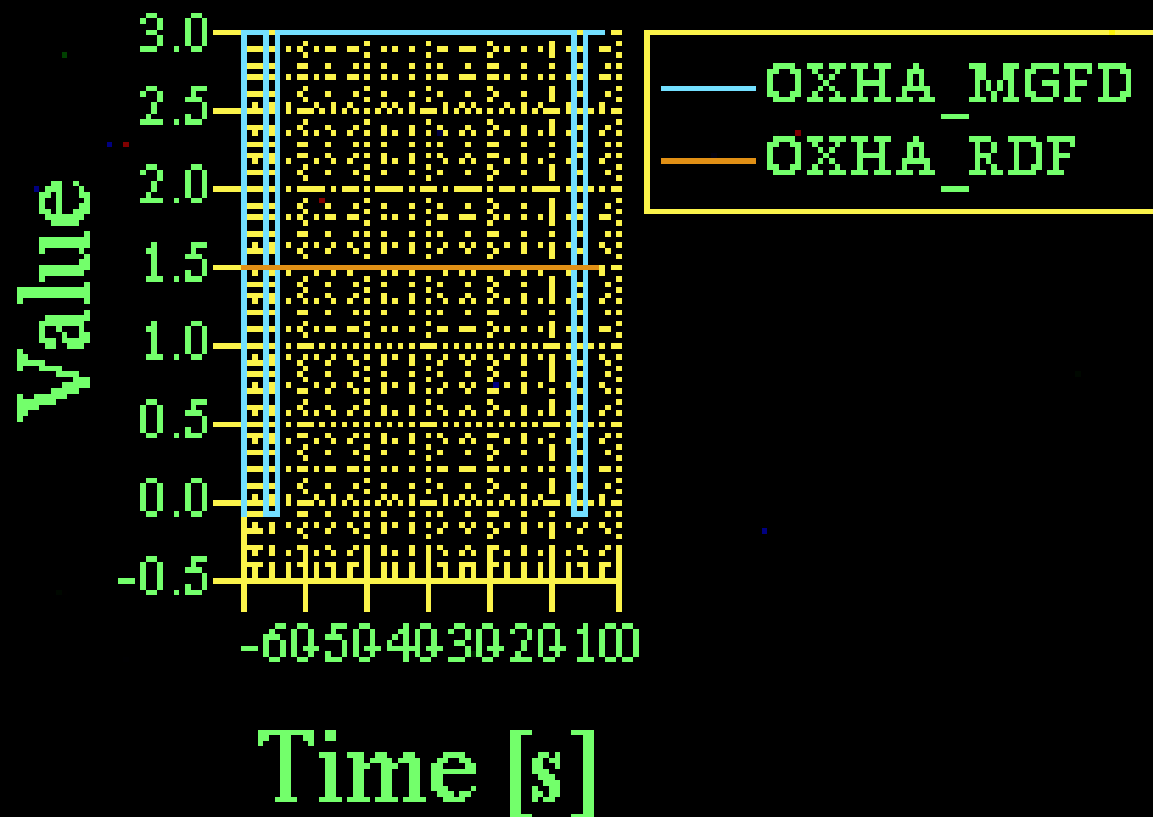


The Møller Magnet Power Supply Screen

EDM(HLA Main) → Moller → Hall A Solenoid

Halla Expert Solenoid

Solenoid Control



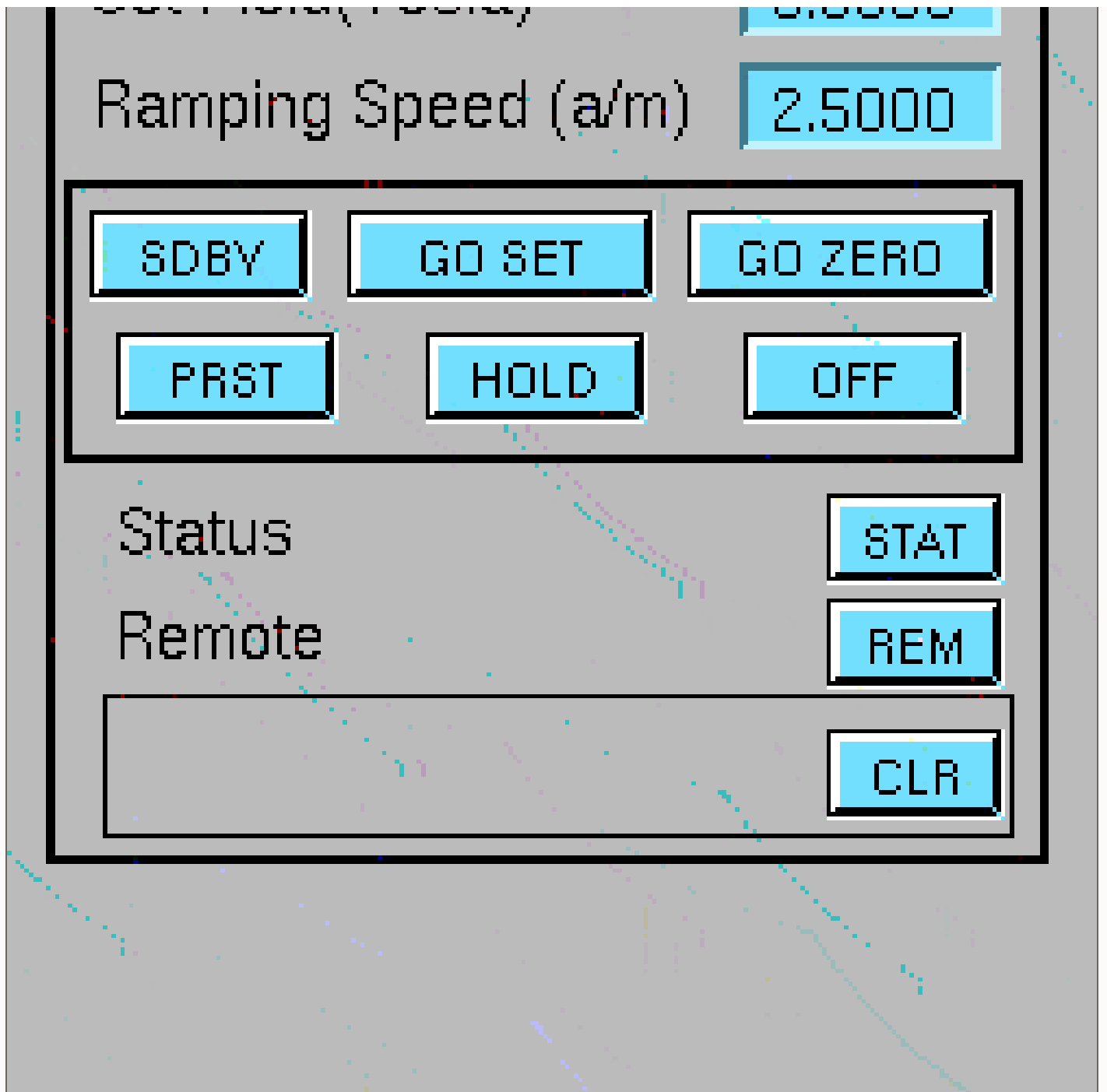
Leads 'field' 1.5009

Persistent field (T) -0.0800

STATUS PERSISTENT

HEATER OFF

Set Field(Tesla) 3.0000



The Møller Magnet Temperature Readbacks Screen

EDM(HLA Main) → Moller → Temperature Readbacks

Hall A Moller Tempature Readbacks	
Temp Probe 1	293.480
Temp Probe 2	292.870
Temp Probe 3	294.710
Temp Probe 4	294.340
Temp Probe 5	0.000
Temp Probe 6	0.000
Temp Probe 7	0.000
Temp Probe 8	126.930

The Møller Magnet Temperature ADCs Screen

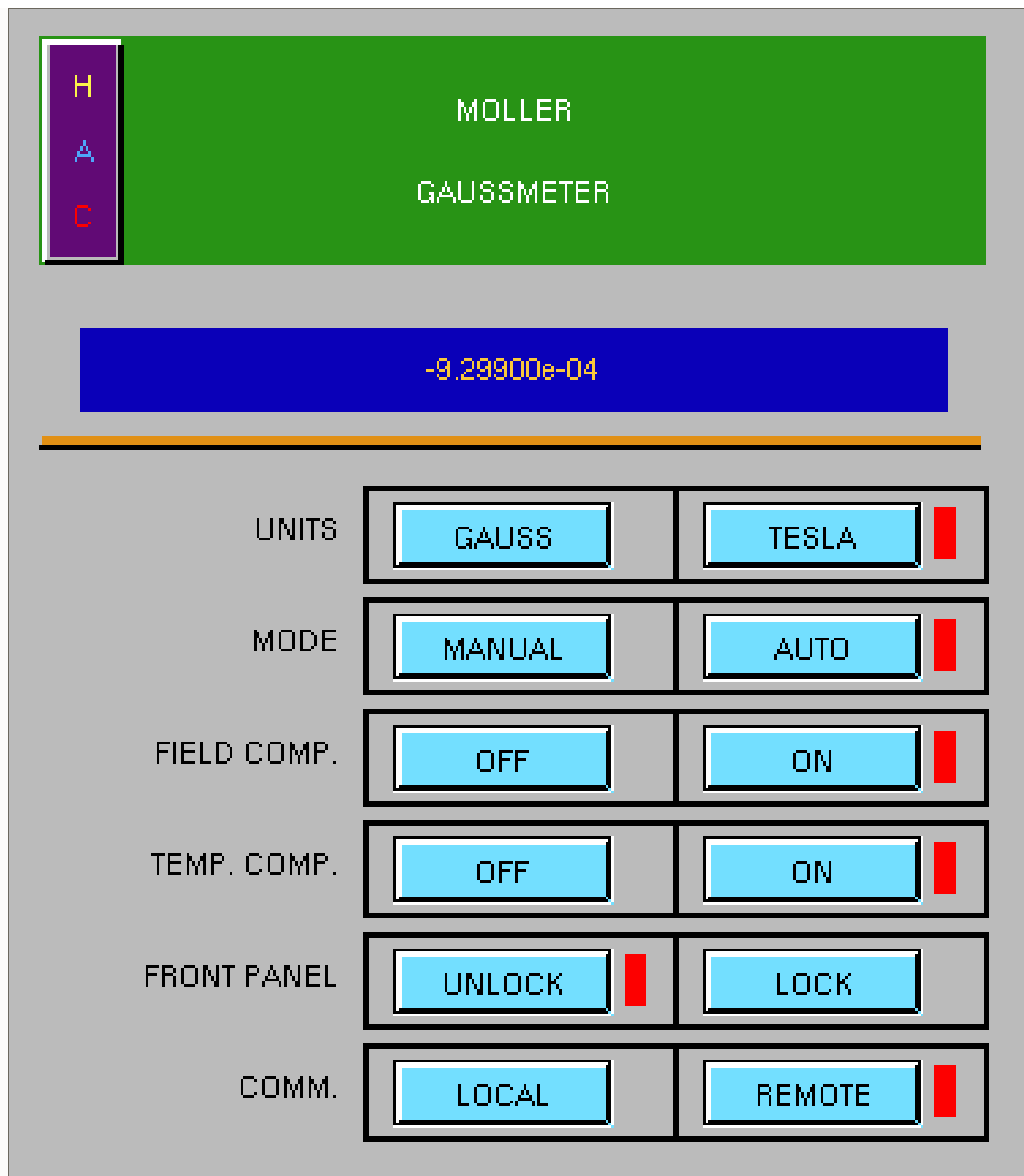
EDM(HLA Main) → Moller → ADCs (VMI3128)

Hall A Moller ADCs

1	-2.224	17	-2.183
2	-2.220	18	-2.174
3	-2.218	19	-2.186
4	-2.260	20	-2.208
5	-2.220	21	-2.152
6	-2.209	22	-2.142
7	-2.211	23	0.000
8	-2.236	24	0.000
9	-2.329	25	0.000
10	-2.327	26	0.000
11	-2.329	27	0.000
12	-2.367	28	0.000
13	-2.328	29	0.000
14	-2.311	30	0.000
15	-2.312	31	0.000
16	-2.345	32	0.000

The Møller Magnet Gaussmeter Screen

EDM(HLA Main) → Moller → Lakeshore Gaussmeter



The Møller Magnet VMIC2210 Screen

EDM(HLA Main) → Møller → VMIC2210

VMIC2210

Ch 0

OFF

ON

Ch 1

OFF

ON

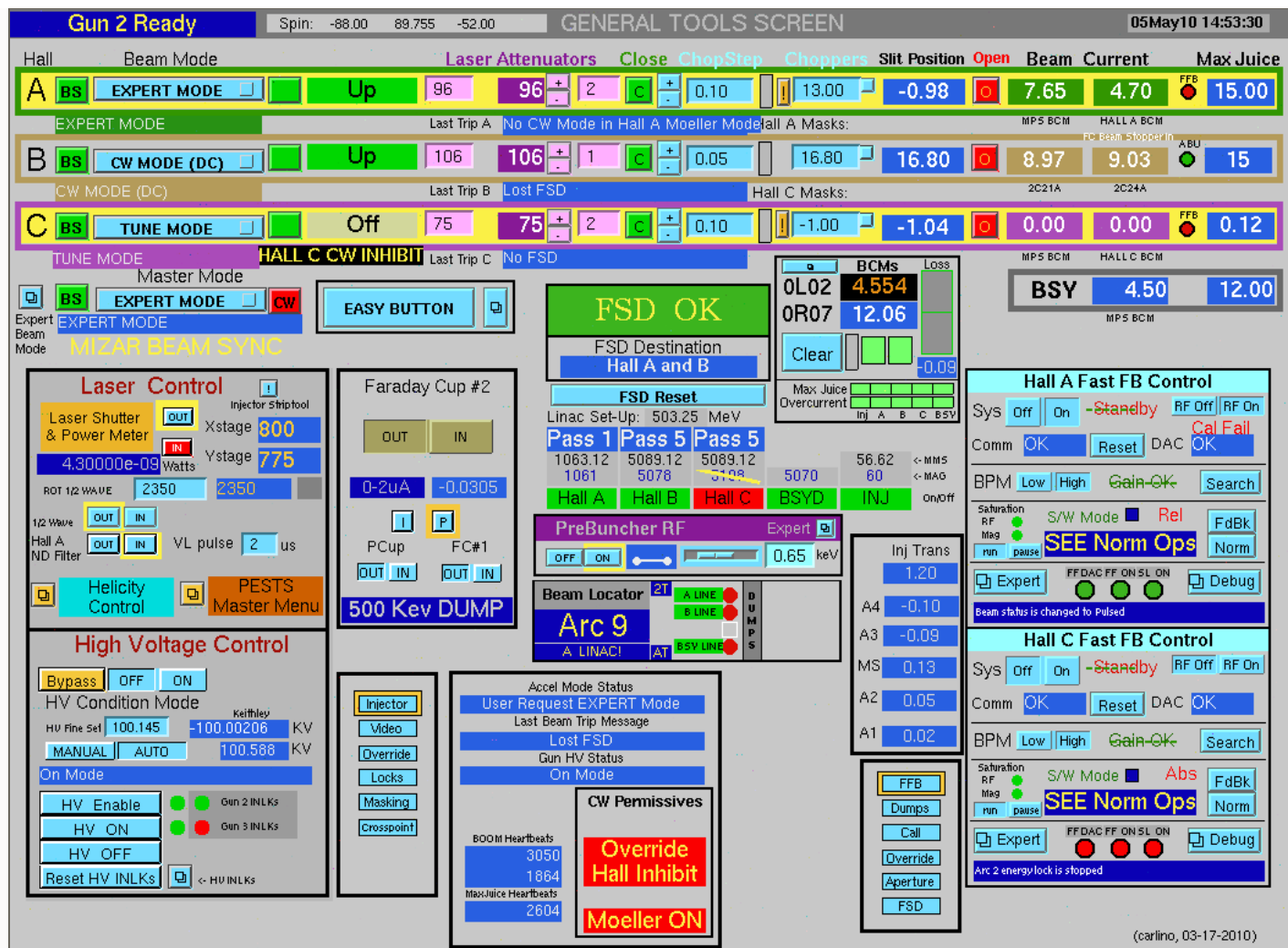
Ch 2

OFF

ON

PGun General Tools Screen

EDM(OPS) → Acc.Main Menu → Tools → General Tools PGUN 3 Laser



Expert Beam Mode Screen

EDM(OPS) → Acc.Main Menu → Tools → General Tools PGUN 3 Laser → Expert Beam Mode

Expert Beam Mode Control

Gun Beam Mode

Source Status

Pol Src Ready

Switch Sources



Select Accel Mode

TUNE MODE



Viewer Not Safe

Accel Mode Status

User Request Tune Mode

Last Beam Trip Message

No FSD

Select A Laser Mode

TUNE MODE



Viewer Not Safe

Hall A Mode Status

User Request Tune Mode

HallA Last Beam Trip Message

No FSD

Select B Laser Mode

BEAM SYNC ONLY



Viewer Safe

Hall B Mode Status

User Request Beam Sync Only

HallB Last Beam Trip Message

Hall B Mode > Accel

Select C Laser Mode

BEAM SYNC ONLY



Viewer Safe

Hall C Mode Status

Beam Sync Only

HallC Last Beam Trip Message

Hall C Mode > Accel

Beam Sync Status

ON LINE SYNC

60 HZ

60 Hz

**Beam Sync ON
SCAM BEAM SYNC**

ON LINE SYNC



Mizar Controls

Viewer Limited Settings

VL Pulse Width Set 2.000 us

VL Pulse Width Read 2.000 us

Modulation Width Read 99.500 us

CW Checks

CW Crew Chief

Permissive

CW ENABLE

CW INHIBIT

CW INHIBIT

Hall A

A ENABLE

A INHIBIT

CW INHIBIT

Hall C

C ENABLE

C INHIBIT

CW ENABLE

Hall B

B ENABLE

B INHIBIT

CW ENABLE

CW MSlit/C.Plug

Permissive



Override ON

Override OFF

CW ENABLE

CW Hall/Slit

Permissive



Override ON

Override OFF

CW INHIBIT

When Expert is selected on Beam Mode beam sync will be set by the Mizar instead of the SCAM. In this mode the Lasers will only allow operation in Expert or Beam Sync modes.

Mizar Controls Screen

EDM(OPS) → Acc.Main Menu → Tools → General Tools PGUN 3 Laser → Expert Beam Mode → Mizar Controls

MIZAR EXPERT GUN CONTROL

Gun Beam Mode

Select Beam Mode

BEAM SYNC ONLY ☐

Beam Mode Status

Beam Sync Mode

Beam Trip Message

Dump Moving or Unknown Stat

Beam Parameters

Source Status

Duty Factor 0.00 %

Pol Src Ready

Switch Sources



Clock and Delay Control

Select Reference Clock

SCAM SYNC ☐

Select Line Sync Frequency

60 HZ ☐

Set Delays

Clock Ref to Beam Sync Delay 0.00 us

Beam Sync To Beam Delay 0.00 us

Beam Sync To Mod Delay 0.00 us

Beam Envelope Control

Viewer Limited Settings

VL Pulse Width 10.00 us

VL Pulse Period 16666.67 us

VL Pulse Frequency 60.00 Hz

Status Setting Accepted

Viewer Not Safe !!!

Pulse Mode Settings

Pulse Width 100.00 us

Pulse Period 16666.67 us

Pulse Frequency 60.00 Hz

Status Setting Accepted

CW Mode Settings

Pulse Width 100.00 us

Pulse Period 16666.67 us

Pulse Frequency 60.00 Hz

DC ON Control OFF ON

Status Init

Modulation Mode Settings

Number of Mod Pulses 1

Modulation Width 0.00 us

Modulation Low 0.00 us

Modulation Period 0.00 us

Modulation Frequency 0.00 Hz

Status Init

Beam Sync Settings

Beam Sync Width 100.00 us

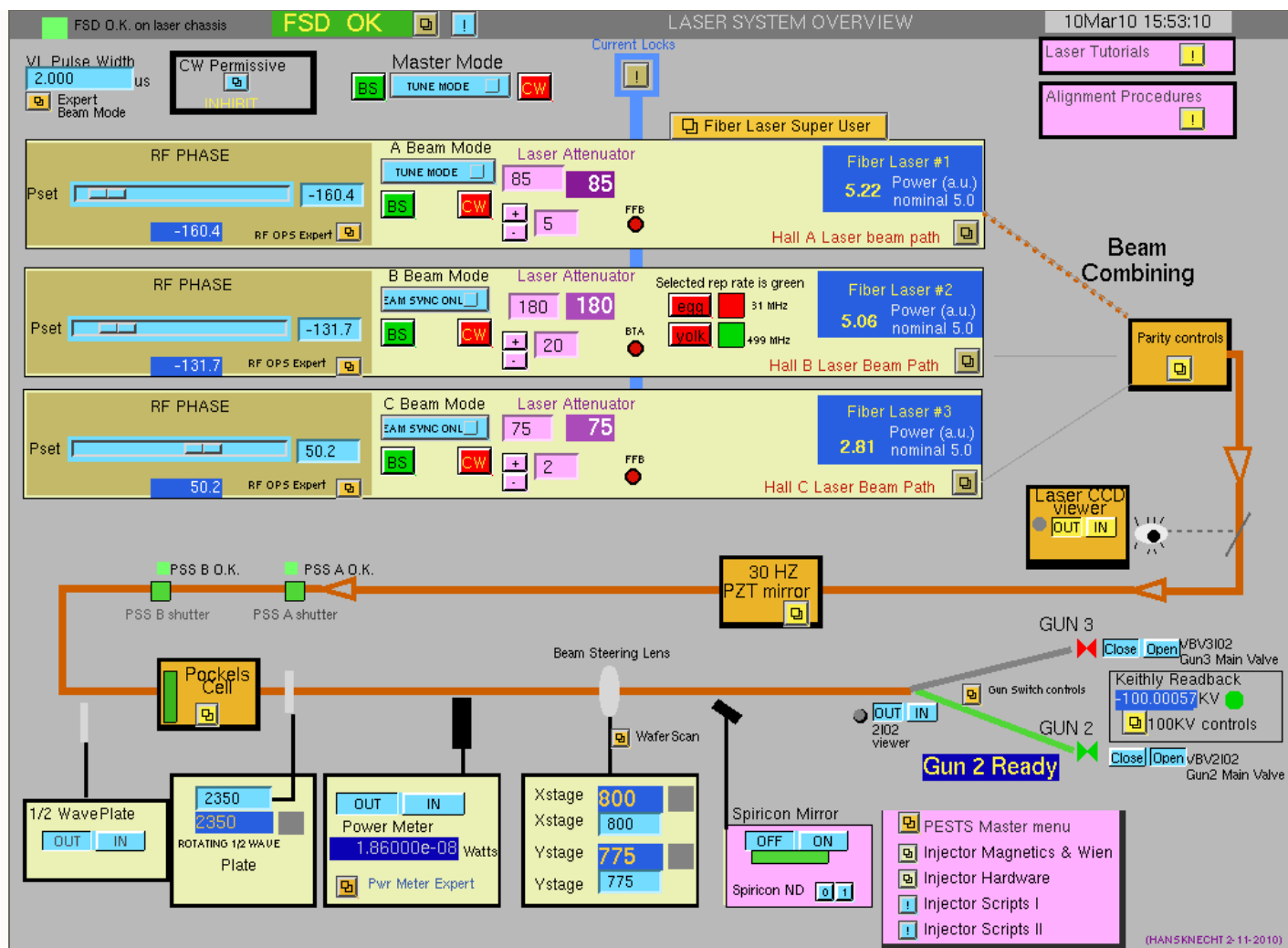
Beam Sync Period 16666.67 us

Beam Sync Frequency 60.00 Hz

Status Setting Accepted

Injector General Tools Screen

EDM(OPS) → Acc.Main Menu → Gun/Laser → PGUN OPS Main



Helicity Control Board Screen

EDM(OPS) → Acc.Main Menu → Gun/Laser → PGUN OPS Main → Helicity Control Board

Helicity Control Board

When Configuration is changed please contact
Scott Higgins to set new configuration as default

	CONTROL	MONITOR
Mode	Free Clock ☐	Free Clock
1 T-SETTLE Input 1	100 usec ☐	100
2 T-SETTLE Input 2		
1 T-STABLE Input 1	4067 usec ☐	4067
2 T-STABLE Input 2		
Helicity Pattern	Octet ☐	Octet
Reporting Delay	16 windows ☐	16 windows
Helicity Board Frequency (Hz)		239.98

Hardware Rev. MONTH **3** DAY **5** YEAR **10**

User Guide **!**

FSD 1H001 Screen

EDM(OPS) → Acc.Main Menu → FSD → FSD Overview → 1H001

1H001 (Normal Card)

UM S R G F

U M

U M

U M

U M

U M

U M

U M

U M

U M

U M

U M

U M

U M

U M

U M

Watchdog Timer - 1H001

BLM 4290:ILM1P01

BLM 4290:ILM1P02

BLM 4290:ILM1P03

BLM 4290:ILM1P04

Vacuum - shield wall to dump

Vacuum - compton chicane

Moeller Target

1H002 Tree

Fast Raster

Fault Trip Reset

Mask Set

56832

Mask Readback

56832

Gold Mask

56832

Latched Fault

8608

Mask Enabled

1

UM = Unmask / Mask

S = Mask Set (grey=unmasked yellow=masked)

R = Mask Readback (grey=unmasked yellow=masked)

G = Gold Mask (grey=unmasked yellow=masked)

F = Latched Fsd Fault (green=good red=fault)

Bit #0 = Top Bit #15 = Bottom

Møller Target FSD Screen

EDM(OPS) → Acc.Main Menu → FSD → FSD Overview → 1H001 → Moller Target

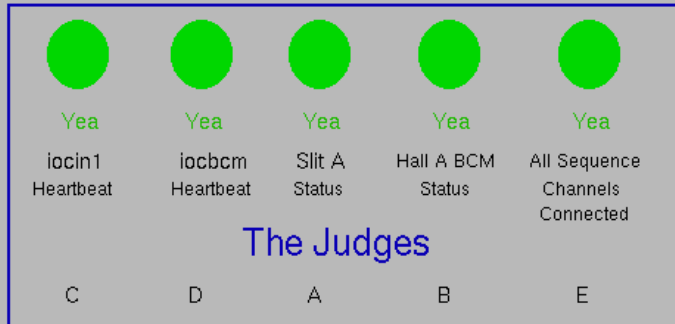
The Question: Should we mask the Hall A Moeller Target FSD?

! Target Mask Request 0 Request Readback
! Target UNMask Request

8.00 Slit Value iocin1:heartbeat 370323
0.00 BLA Current iocbcm:heartbeat 3476833

ISD1H001L.H
(Moeller Mask Request)

ISD1H001NA.B7
(Moeller Mask Hardware Readback)



Application is fsdwft (wft = waterfall target) -- also see the fsd application.

The Moeller Target can be moved without an FSD trip, only if the Moeller FSD is masked.

The Moeller FSD can be masked only from this screen via the above [I] Target Mask Request button.

Masking is allowed only when all 5 Judges are green (Yea).

The Slit A Judge is green only for a slit position of 8. [gel:19mar2010]

The Hall A BCM Judge is green only for a beam current of less than 0.5 uA. [gel:19mar2010]

BPM Summary Screen

EDM(OPS) → Acc.Main Menu → BPM → BPM Spikes and Position Summary → Position Summary → Hall A Transport

Hall A Position Summary											
SEE AG Sel	BPM	X Position Posset	Y Position Posset	X Position Absolute	Y Position Absolute	X Position Relative	Y Position Relative	X Position RMS	Y Position RMS	Zero Pos Time Stamp	
Auto Select											
STP	IPM1C00	-0.178	1.928	-0.217	1.845	-0.040	-0.082	0.013	0.008	04JUN10 10:26:08	
STP	IPM1C01	1.262	-0.087	1.296	-0.314	0.034	-0.227	0.013	0.012	04JUN10 10:26:08	
STP	IPM1C02	1.120	1.812	1.171	1.414	0.052	-0.398	0.008	0.022	04JUN10 10:26:08	
STP	IPM1C03	-0.065	1.151	0.016	0.866	0.081	-0.284	0.008	0.013	04JUN10 10:26:08	
STP	IPM1C04	0.961	-0.049	1.028	-0.200	0.067	-0.151	0.013	0.008	04JUN10 10:26:08	
STP	IPM1C05	0.971	0.114	1.019	-0.063	0.048	-0.177	0.010	0.008	04JUN10 10:26:08	
STP	IPM1C06	1.917	-0.073	1.939	-0.174	0.022	-0.101	0.019	0.005	04JUN10 10:26:08	
SLN	IPM1C07	No ZPOS	No ZPOS	-0.128	0.024	-0.128	0.024	0.030	0.058	04JUN10 10:26:20	
SLN	IPM1C08	0.000	0.000	0.004	-0.093	0.004	-0.093	0.039	0.055	04JUN10 10:26:20	
STP	IPM1C10	0.395	0.039	0.419	0.094	0.023	0.055	0.005	0.015	04JUN10 10:26:08	
SLN	IPM1C11	0.000	0.000	-0.165	-0.010	-0.165	-0.010	0.039	0.066	04JUN10 10:26:20	
SLN	IPM1C12	0.000	0.000	-0.013	0.015	-0.013	0.015	0.057	0.049	04JUN10 10:26:20	
SLN	IPM1C14	0.000	0.000	-0.121	-0.365	-0.121	-0.365	0.034	0.053	04JUN10 10:26:20	
SLN	IPM1C16	0.000	0.000	-0.016	0.035	-0.016	0.035	0.098	0.053	04JUN10 10:26:20	
SLN	IPM1C18	0.000	0.000	0.003	0.151	0.003	0.151	0.079	0.056	04JUN10 10:26:20	
SLN	IPM1C20	0.000	0.000	-0.094	-0.142	-0.094	-0.142	0.046	0.048	04JUN10 10:26:20	
SLN	NP IPM1P02A	No ZPOS	No ZPOS	0.000	0.000	0.000	0.000	0.000	0.000	04JUN10 10:26:07	
SLN	NP IPM1P02B	No ZPOS	No ZPOS	0.000	0.000	0.000	0.000	0.000	0.000	04JUN10 10:26:07	
SLN	NP IPM1P03A	No ZPOS	No ZPOS	0.000	0.000	0.000	0.000	0.000	0.000	04JUN10 10:26:07	
SLN	IPM1H01	0.807	0.353	0.794	0.376	-0.013	0.023	0.014	0.027	04JUN10 10:26:07	
SLN	IPM1H04A	-0.174	-0.072	-0.222	-0.060	-0.048	0.012	0.022	0.053	04JUN10 10:26:07	
SLN	IPM1H04B	0.072	1.007	0.047	1.082	-0.026	0.075	0.025	0.067	04JUN10 10:26:07	

Hall A Absolute Spikes

EDM(OPS) → Acc.Main Menu → BPM → BPM Overview → Hall A Absolute

CW MODE (DC)

Hall A Absolute Spikes

(screen generated by EDMPlus,
Ilaps BPM team, May 26 2010)

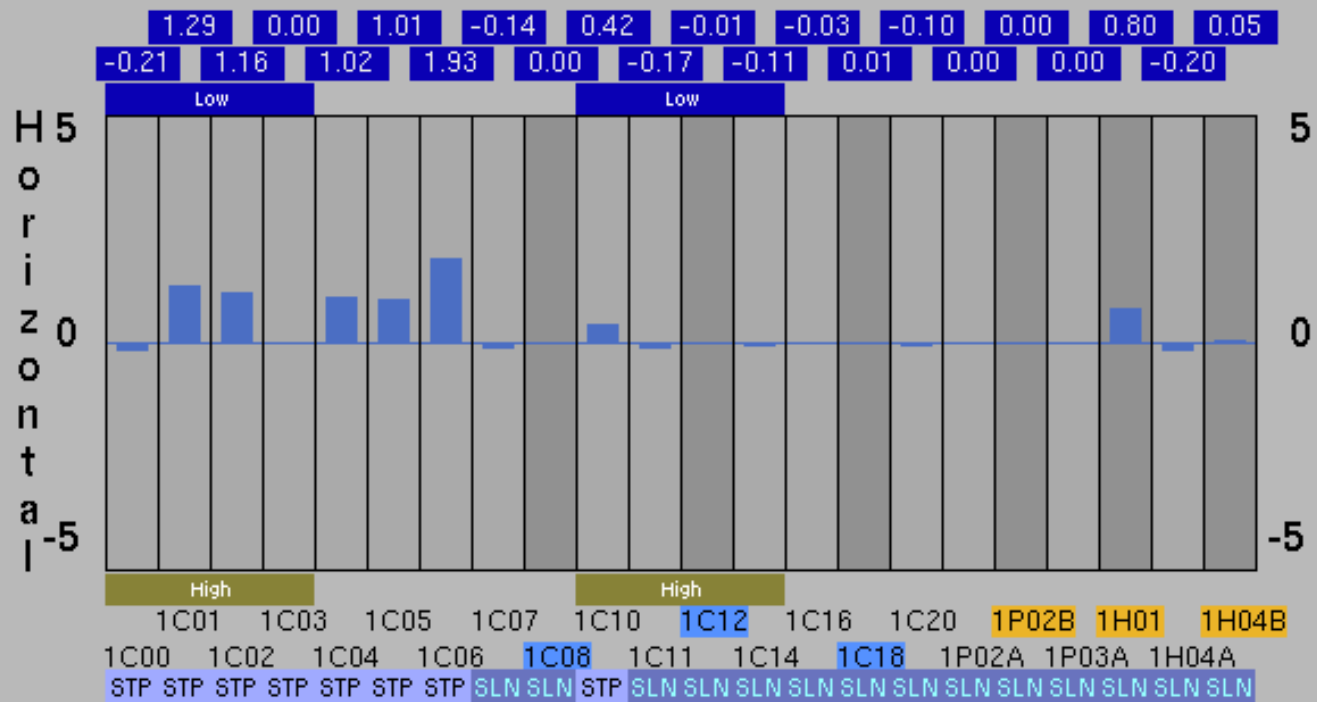
SCRAPER IN

04Jun10 14:17:2

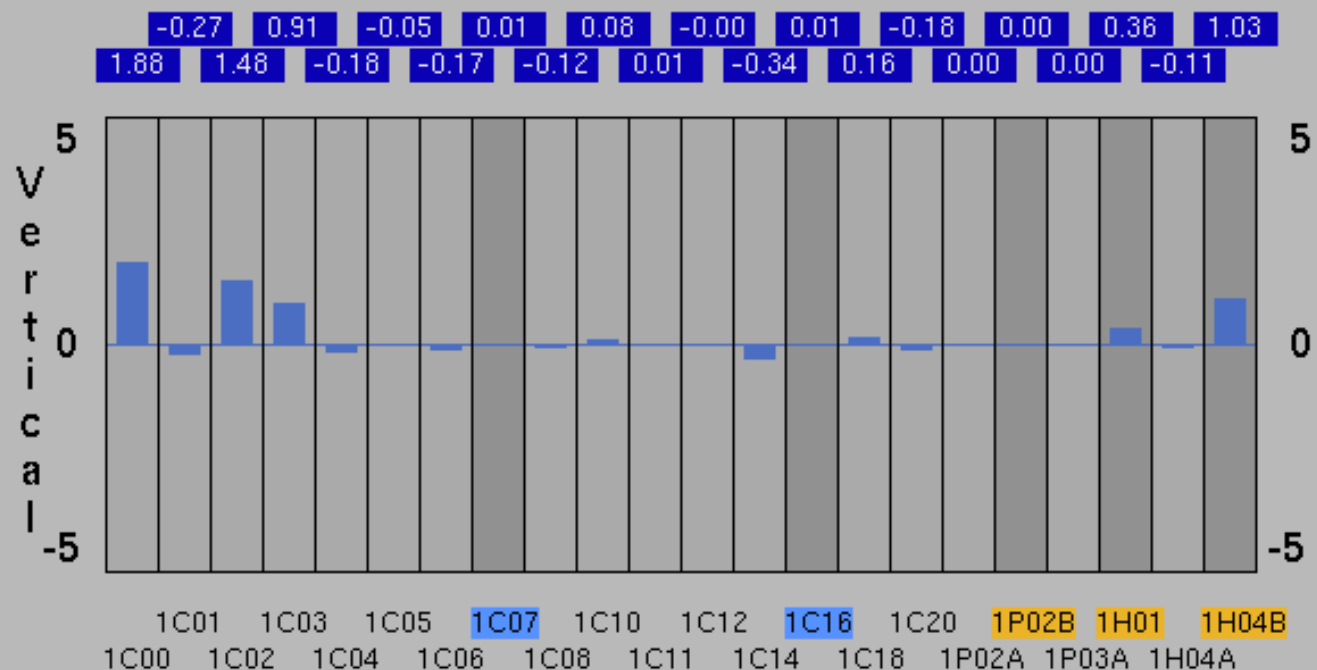
Utilities

iocse8
iocse9
iocse10SEE Nom Ops On
Feedback On
SEE Nom Ops Off

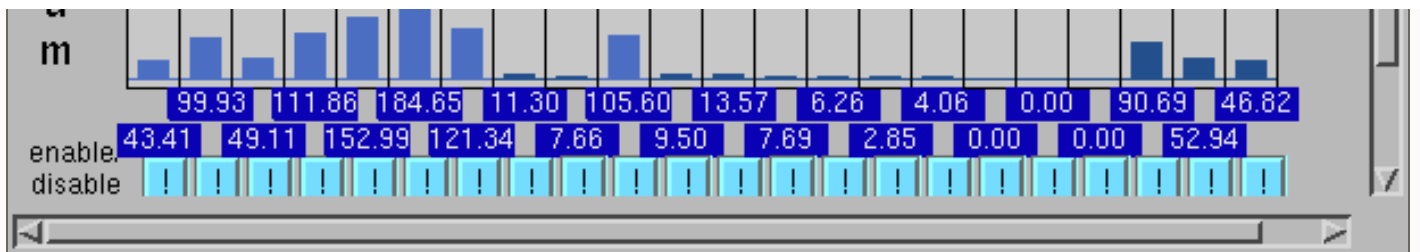
Rel Spikes

POS ☐

NP NP NP



WS



Beam Energy Monitor

EDM(OPS) → Acc.Main Menu → Tools → Beam Energy Monitor

Beam Energy Monitor							
off/on							server heartbeat 103520
Master	off/on	dp/p bpm	dp/p corr	dp/p total	MeV	Target MeV	Status Invalid
INJ	off/on	2.657e-04	1.721e-02	1.748e-02	57.161	56.616	POS/BEAM
ARC1	off/on	-9.399e-05	1.953e-04	1.013e-04	564.171	559.866	Beam Here
ARC2	off/on	0.000e+00	0.000e+00	0.000e+00	1061.581	1063.116	NO BEAM
ARC3	off/on	0.000e+00	0.000e+00	0.000e+00	1573.123	1566.366	No Beam
ARC4	off/on	0.000e+00	0.000e+00	0.000e+00	2076.575	2069.616	NO BEAM
ARC5	off/on	0.000e+00	0.000e+00	0.000e+00	2582.844	2572.866	No Beam
ARC6	off/on	0.000e+00	0.000e+00	0.000e+00	3074.877	3076.116	POS/BEAM
ARC7	off/on	0.000e+00	0.000e+00	0.000e+00	3595.436	3579.366	No Beam
ARC8	off/on	0.000e+00	0.000e+00	0.000e+00	4080.695	4082.616	POS/BEAM
ARC9	off/on	0.000e+00	0.000e+00	0.000e+00	4595.207	4585.866	No Beam
HALLA	off/on	8.595e-08	0.000e+00	8.595e-08	1060.524	1063.116	POS/BEAM
HALLC	off/on	0.000e+00	0.000e+00	0.000e+00	5092.652	5089.116	NO BEAM

(carlo, November 09 2009)